Rakan Abdulla

Mechatronic Engineer | Enthusiastic about Mine Automation & Sustainability

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Education

University of Sydney — Bachelor of Engineering (Honours)
Mechatronic Engineering, 2022 – Present
- Current WAM: 75%
- Relevant Coursework: Control Systems, Robotics, Embedded Systems, CAD, Manufacturing Engineering
- Hackathon Participant and Project Leader (see below)

Project Experience

RoboDog – Assistive Robot for People with Allergies & Disabilities (2024)
- Developed a Bluetooth-controlled robot using an FPGA board to follow, respond, and dance on command.
- Integrated sensors and control systems, reflecting real-world mine automation systems.

Autonomous Shopping Cart with TurtleBot and ROS2 (2024)
- Programmed an autonomous TurtleBot using ROS2 to identify, count, and restock items using QR codes.
- Applied logic and pathfinding applicable to warehouse and mining automation.

Satellite Refueling CAD Design (ANT61) (2024)
- Designed a prototype latching mechanism for autonomous satellite refueling using Fusion360.
- Demonstrated mechanical design and problem-solving in a high-precision context.

Conference Carbon Calculator (Team Leader) (2024)
- Led a 4-person team to develop a tool that estimates and reduces emissions for events.
- Applied sustainability principles, relevant to South32's ESG goals.

Whack-A-Mole Game on FPGA (2024)
- Built a hardware-based reaction game with multiple difficulty modes using Verilog and FPGA.

Technical Skills

Software: SolidWorks, Fusion360, MATLAB, Excel, Word, PowerPoint
Programming: Python, C, C++, Arduino, Assembly, Verilog, R
Platforms: ROS2, FPGA, Embedded Systems
Skills: Automation, Robotics, Mechanical Design, Control Systems

Transferable Skills

Leadership: Successfully led technical teams from planning to delivery.
Collaboration: Worked across teams in both industry (CloudAk, Escape Room) and uni projects.
Communication: Experienced in client interaction and technical presentations.
Adaptability: Learned ROS2, FPGA workflows, and CAD tools independently.

Industry Experience

CloudAk — Testing Technician Intern
May 2021 – August 2021
- Supported server system testing and maintenance in data center environments.

Escape The Room — Escape Room Engineer Intern
September 2021 – November 2021
- Designed mechanical puzzles and implemented engineering concepts in real-world client products.

References

Available on request.